Cambridge
O Level

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Cambridge Ordinary Level

CHEMISTRY **5070/22**

Paper 2 Theory **May/June 2017**

MARK SCHEME

Maximum Mark: 75

Published

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Question	Answer	Mark
A1(a)	Copper(II) chloride	**1**
A1(b)	Ammonium chloride	**1**
A1(c)	Hydrogen chloride	**1**
A1(d)	Ammonium chloride	**1**
A1(e)	Carbon tetrachloride	**1**

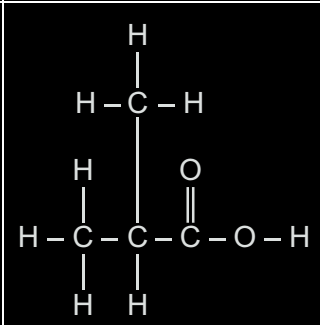
Question	Answer	Mark
A2(a)	<table><tr><th>particle</th><th>atomic number</th><th>number of neutrons in particle</th><th>number of electrons in particle</th></tr><tr><td>^{35}Cl</td><td>17</td><td>18</td><td>**17** (1)</td></tr><tr><td>**^{37}Cl** (1)</td><td>17</td><td>20</td><td>17</td></tr><tr><td>$^{39}K^{+}$</td><td>19</td><td>**20** (1)</td><td>18</td></tr><tr><td>$^{79}Br^{-}$</td><td>**35** (1)</td><td>44</td><td>36</td></tr><tr><td>^{81}Br</td><td>35</td><td>**46** (1)</td><td>35</td></tr><tr><td>**$^{85}Rb^{+}$** (1)</td><td>37</td><td>48</td><td>36</td></tr></table>	**6**
A2(b)(i)	Atoms with same number of protons but different number of neutrons / atoms with same atomic number but different nucleon number / atoms of the same element with different number of neutrons	**1**
A2(b)(ii)	^{35}Cl and ^{37}Cl	**1**

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Question	Answer	Mark
A3(a)(i)	Hydrochloric acid **AND** magnesium oxide	**1**
A3(a)(ii)	1 mark each for any 4 of: • Use of **excess** base (1) • Use hot acid / use warm acid / warm the mixture (of acid and base) (1) • Filter mixture (to get filtrate) (1) • Evaporate some of filtrate and allow to crystallise / leave in warm place to crystallise / heat to crystallisation point (1) • (Filter), wash with organic solvent / dry with filter paper / dry in a (drying) oven (1)	**4**
A3(b)	$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$ Correct formulae and balancing (1) Correct state symbols – dependent on correct formulae (1)	**2**
A3(c)(i)	Moles of acid = 0.020×0.65 **OR** 0.013 (1) Mass = 2.26(2) (g) / 2.3 (g) (1)	**2**
A3(c)(ii)	Percentage yield = 76.(1) %	**1**

Question	Answer	Mark
A4(a)	Sodium ion: 2.8 (1) Oxide ion: 2.8 (1)	**2**
A(b)	Negative electrode: $Na^+ + e^- \rightarrow Na$ (1) Positive electrode: $2O^{2-} \rightarrow O_2 + 4e^-$ (1)	**2**
A(c)	Ions move / mobile ions / ions free to move	**1**
A(d)	$Na_2O + H_2O \rightarrow 2NaOH$	**1**

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Question	Answer	Mark
A5(a)	(Acidified) potassium manganate(VII) / oxygen	**1**
A5(b)(i)	Lithium / sodium / potassium / calcium / magnesium (1) Corresponding ethanoate **AND** hydrogen (1)	**2**
A5(b)(ii)	$CaCO_3 + 2CH_3CO_2H \rightarrow Ca(CH_3CO_2)_2 + H_2O + CO_2$ (2) IF: two marks not scored H_2O and CO_2 as products = 1 mark	**2**
A5(c)		**1**
A5(d)(i)	Condensation	**1**
A5(d)(ii)	Decomposes / decays / will not fill up land-fill sites / less litter / no need for incineration	**1**

Question	Answer	Mark
A6(a)	Energy / enthalpy on vertical axis **AND** progress of reaction / course of reaction on horizontal axis (1) Reactant level above product level and to the left of product **AND** reactants and products labelled (1) Enthalpy change shown by downward arrow **AND** labelled enthalpy change or ΔH (1)	**3**
A6(b)	1 mark each for any **two** of: • Lower activation energy (1) • More particles have energy equal to / greater than the activation energy (1) • Different pathway / different mechanism / via an enzyme complex (1) • more successful collisions (between groups on enzyme and substrates) / number of effective collisions increase (with specific groups on enzyme surface) (1)	**2**
A6(c)	Idea that combustion **AND** respiration increase levels of carbon dioxide / carbon in the atmosphere (1) Idea that photosynthesis reduces levels of carbon dioxide / carbon in the atmosphere (1) Idea that these processes balance each other (1)	**3**

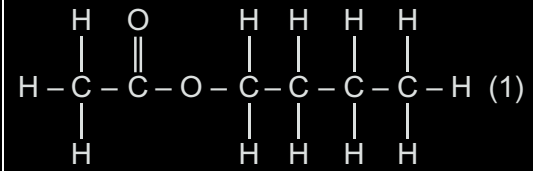
Question	Answer	Mark
B7(a)	Blue solution / bubbles	**1**
B7(b)(i)	Copper(II) sulfate	**1**
B7(b)(ii)	Copper loses electron(s)	**1**
B7(c)	Moles of acid = 0.025×14.0 **OR** 0.35 (1) Moles of sulfur dioxide = 0.175 (1) Volume of gas = 4.2 dm^3 / 4 200 cm^3 (1)	**3**
B7(d)(i)	Blue precipitate / blue solid (which does not redissolve)	**1**
B7(d)(ii)	Blue precipitate / blue solid (1) In excess ammonia gives a dark blue solution (1)	**2**
B7(e)	$2CuCl \rightarrow CuCl_2 + Cu$	**1**

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Question	Answer	Mark
B8(a)	Reversible reaction	**1**
B8(b)	(Reaction in which) heat is released	**1**
B8(c)	Idea that no reactants or products can enter / leave	**1**
B8(d)	The colour becomes more brown / colour becomes darker (1) Fewer moles on right hand side so position of equilibrium moves to the left (or reverse argument) / fewer moles on product side so position of equilibrium moves to the left (1)	**2**
B8(e)	The colour more brown / colour becomes darker (1) Exothermic reaction so position of equilibrium moves to the left / backward reaction is endothermic so equilibrium moves to left(1)	**2**
B8(f)(i)	• Add a reactive metal / carbonate to the two acids at the same concentration (1) AND 1 mark for any one of: • Time how long it takes for the metal / carbonate to disappear (1) • Time how long it takes to produce a fixed volume of gas (1) • Count the number of bubbles over fixed time interval (1) • Weak acid has a long**er** reaction time (or reverse argument) / weak acid produces few**er** bubbles in a given time interval (1) **OR** • Add universal indicator to the two acids at the same concentration (1) AND 1 mark for either one of: • Compare colour with colour chart (1) • Red with strong acid AND yellow with weak acid (1) **OR** • Dip pH meter into the two acids at the same concentration (1) AND 1 mark for either one of: • Record pH (1) • pH low**er** for strong acid / pH less for strong acid (than for weak acid) (or reverse argument) (1)	**2**
B8(f)(ii)	KNO_2 **AND** KNO_3	**1**

Question	Answer	Mark
B9(a)	Fuel	**1**
B9(b)	Decomposing vegetation	**1**
B9(c)	Climate change / global warming	**1**
B9(d)(i)	H C C*l* 0.040 0.242 0.718 **OR** 1 12 35.5 0.040 mol 0.020 mol 0.020 mol (1) $CH_2C\mathit{l}$ (1)	**2**
B9(d)(ii)	$C_2H_4C\mathit{l}_2$	**1**
B9(e)(i)	The (overall) movement of particles from high concentration to a low concentration / mixing due to (random) movement of particles	**1**
B9(e)(ii)	Particles are moving faster / particles have more kinetic energy	**1**
B9(e)(iii)	**Molecules**/**particles** have different (relative formula) masses / **molecules**/**particles** have different (relative molecular) masses (1) Methane (molecules) move or diffuse faster / butane (molecules) move or diffuse more slowly (1)	**2**

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Question	Answer	Mark
B10(a)(i)	$C_nH_{2n+1}OH$ / $C_nH_{2n+2}O$	**1**
B10(a)(ii)	Any value between 154 – 164 (°C) (inclusive of these values)	**1**
B10(b)	(Add) yeast (1) Temperature between 5 and 40 °C / no oxygen present / anaerobic (1) (Fractionally) distil (to get ethanol) (1)	**3**
B10(c)	Butyl ethanoate (1)  (1)	**2**
B10(d)	They get slower / they move less rapidly (when temperature decreases) / molecules slow down (when temperature decreases) / molecules have less kinetic energy (when temperature decreases) (1) They / molecules get closer together (when temperature decreases) (1) They / molecules arranged less randomly / less irregularly (when temperature decreases) (1)	**3**

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